

DATA ANALYSIS PROJECT ROADMAP

BY KESHDANALYST

A practical step-by-step framework for approaching any project projects from problem definition to final communication.

From a Question → to Evidence → to a Meaningful Conclusion.

THE COMPLETE ROADMAP

Use this as the mental map for every project.

01 Problem

02 Aim

03 Objectives

04 Questions

05 Data
Requirements

06 Collection

07 Explore

08 Clean

09 Transform

10 Analyze

11 Visualize

12 Insights

13 Recommendation
s

14 Conclusion

15 Report

16 Document

Think in sequence — but revisit earlier steps when new evidence changes your understanding.

01 PROBLEM STATEMENT

What problem, challenge, gap, or situation needs investigation?

01

KEY IDEA

What problem, challenge, gap, or situation needs investigation?

ASK → INVESTIGATE

DEFINE THE SITUATION

Describe what is happening or what needs attention.

DEFINE THE GAP

State what is unknown, missing, inefficient, or unclear.

DEFINE THE IMPACT

Explain why solving or understanding it matters.

EXAMPLE / APPLICATION

Example: "Participation in the programme has fallen, but the reasons are unclear."

ANALYST / PROJECT TIP

A strong problem statement is specific enough to investigate — but broad enough to allow evidence to guide the answer.

02 PROJECT AIM

What is the overall purpose of the project?

02

KEY IDEA

What is the overall purpose of the project?

ASK → INVESTIGATE

ONE OVERALL PURPOSE

State the main thing the project intends to achieve.

KEEP IT BROAD

The aim gives direction; it does not list every task.

MAKE IT ALIGNED

It should directly address the problem statement.

EXAMPLE / APPLICATION

Example: "To investigate the factors associated with programme participation."

ANALYST / PROJECT TIP

Aim = the destination. Objectives = the work you will do to get there.

03 OBJECTIVES

What specifically will the project investigate or accomplish?

03

KEY IDEA

**What
specifically will
the project
investigate or
accomplish?**

ASK → INVESTIGATE

3-6 OBJECTIVES

Break the aim into manageable pieces.

MAKE THEM MEASURABLE

Use action verbs: analyse, compare, identify, assess, determine.

KEEP THEM RELEVANT

Every objective should help answer the problem.

EXAMPLE / APPLICATION

Example: Analyse participation trends •
Compare groups • Identify major patterns
• Assess possible factors.

ANALYST / PROJECT TIP

If an objective cannot be investigated with your available evidence, revise it before collecting data.

04 KEY QUESTIONS

What do we need the project to answer?

04

KEY IDEA

What do we
need the
project to
answer?

ASK → INVESTIGATE

TURN OBJECTIVES INTO QUESTIONS

Each objective should produce one or more answerable questions.

BE SPECIFIC

Define who/what, when, where, and what measure matters.

PLAN THE EVIDENCE

Know what result would count as a useful answer.

EXAMPLE / APPLICATION

Objective: Compare participation by group
→ Question: "Which group has the highest participation rate?"

ANALYST / PROJECT TIP

Question → evidence → analysis → answer.
This chain keeps the project focused.

05 DATA REQUIREMENTS

What information is needed to answer the questions?

05

KEY IDEA

What information is needed to answer the questions?

ASK → INVESTIGATE

FIELDS / VARIABLES

List the columns or attributes required.

RECORDS / SAMPLE

Define who or what should be included.

TIME & SCOPE

Specify period, location, population, or boundaries.

EXAMPLE / APPLICATION

Possible fields: Date • Category • Location • Participant type • Outcome • Score • Status

ANALYST / PROJECT TIP

Do not collect data simply because it is available. Collect what helps answer the questions.

06 DATA COLLECTION

Where will the data come from?

06

KEY IDEA

**Where will the
data come
from?**

ASK → INVESTIGATE

IDENTIFY SOURCES

Databases, files, surveys, APIs, observations, public datasets, records, etc.

CHECK QUALITY

Consider accuracy, completeness, relevance, and credibility.

RECORD PROVENANCE

Keep the source, date obtained, and important collection details.

EXAMPLE / APPLICATION

A project can combine multiple sources — but the sources must be compatible and clearly documented.

ANALYST / PROJECT TIP

Reliable conclusions begin with appropriate and trustworthy data.

07 DATA UNDERSTANDING / EXPLORATION

What does the dataset actually contain?

07

KEY IDEA

What does the dataset actually contain?

ASK → INVESTIGATE

PROFILE THE DATA

Rows, columns, data types, unique values, ranges.

CHECK DATA QUALITY

Missing values, duplicates, unusual values, inconsistent categories.

LOOK FOR EARLY PATTERNS

Basic counts, distributions, trends, and relationships.

EXAMPLE / APPLICATION

Before analysis, ask: "What is one row? What does each column mean? What does the dataset NOT contain?"

ANALYST / PROJECT TIP

Never assume the dataset means what its column names suggest — inspect and understand it first.

08 DATA CLEANING

Can we trust the data enough to analyze it?

08

KEY IDEA

Can we trust
the data
enough to
analyze it?

ASK → INVESTIGATE

MISSING VALUES

Investigate why values are missing; remove, replace, flag, or retain appropriately.

DUPLICATES & ERRORS

Detect repeated records and impossible or inconsistent entries.

STANDARDIZE

Fix formats, categories, dates, spelling, and data types.

EXAMPLE / APPLICATION

Also investigate outliers and unusual records. Document important cleaning decisions.

ANALYST / PROJECT TIP

Cleaning is not about making data look perfect — it is about making it suitable and defensible for analysis.

09 DATA TRANSFORMATION

Can we make the data more useful for the questions?

09

KEY IDEA

Can we make the data more useful for the questions?

ASK → INVESTIGATE

CREATE VARIABLES

Calculated fields, categories, indicators, ratios, or scores.

CREATE TIME FEATURES

Month, quarter, year, duration, age, etc.

PREPARE FOR ANALYSIS

Reshape, aggregate, merge, split, or derive fields when needed.

EXAMPLE / APPLICATION

Raw: Date of Event → Month / Quarter / Year. Raw: Score 0–100 → Performance Band.

ANALYST / PROJECT TIP

Only transform data when the new structure or variable supports a clear analytical purpose.

10 DATA ANALYSIS

What does the evidence tell us?

10

KEY IDEA

What does the evidence tell us?

ASK → INVESTIGATE

DESCRIBE

Counts, totals, averages, minimums, maximums, distributions.

COMPARE

Groups, categories, periods, locations, or conditions.

INVESTIGATE

Trends, relationships, differences, patterns, and anomalies.

EXAMPLE / APPLICATION

Useful techniques: filtering • sorting • grouping • aggregation • pivot tables • statistics • SQL • trend analysis

ANALYST / PROJECT TIP

Choose the method that best answers the question — not the method that is simply available.

11 DATA VISUALIZATION

How can we communicate the findings clearly?

11

KEY IDEA

How can we
communicate
the findings
clearly?

ASK → INVESTIGATE

MATCH VISUAL TO QUESTION

Bar = comparison • Line = trend • KPI = key number • Scatter = relationship.

REDUCE CLUTTER

Use clear labels, useful scales, and meaningful titles.

HIGHLIGHT THE MESSAGE

Make the important pattern easy to see.

EXAMPLE / APPLICATION

Good title: "Participation increased steadily after the programme redesign."

ANALYST / PROJECT TIP

Do not create charts just because they look attractive. Every visual should communicate something useful.

12 INSIGHTS / FINDINGS

What did we discover — and what does it mean?

12

KEY IDEA

What did we discover — and what does it mean?

ASK → INVESTIGATE

STATE THE FINDING

Describe the important pattern, difference, trend, or anomaly.

INTERPRET IT

Explain why the finding matters in the context of the project.

STAY EVIDENCE-BASED

Separate what the data shows from what you assume.

EXAMPLE / APPLICATION

Data: Group A = 72%, Group B = 48% →
Finding: Group A participation is substantially higher.

ANALYST / PROJECT TIP

Move beyond “what happened?” to “what does this mean?”

13 RECOMMENDATIONS / ACTIONS

What should happen next based on the findings?

13

KEY IDEA

What should happen next based on the findings?

ASK → INVESTIGATE

CONNECT TO INSIGHTS

Each action should trace back to evidence.

BE SPECIFIC

State what should be done, by whom or where appropriate.

CONSIDER FEASIBILITY

Think about resources, risks, limitations, and context.

EXAMPLE / APPLICATION

Insight: Participation is lowest during late sessions → Action: Review timing and test an earlier schedule.

ANALYST / PROJECT TIP

Recommendations should be evidence-informed — not guesses presented as facts.

14 CONCLUSION

What is the overall answer from the project?

14

KEY IDEA

What is the overall answer from the project?

ASK → INVESTIGATE

RETURN TO THE PROBLEM

Remind the audience what was investigated.

SUMMARIZE THE MAIN FINDINGS

Include only the most important results.

STATE THE OVERALL TAKEAWAY

Explain what the project ultimately revealed.

EXAMPLE / APPLICATION

A strong conclusion connects: Problem → Evidence → Main finding → Overall takeaway.

ANALYST / PROJECT TIP

Do not introduce major new evidence in the conclusion.

15 PRESENTATION / REPORTING

How will we communicate the project clearly?

15

KEY IDEA

How will we
communicate
the project
clearly?

ASK → INVESTIGATE

TELL A LOGICAL STORY

Problem → Objectives → Data → Cleaning →
Analysis → Visuals → Insights → Actions →
Conclusion.

FOCUS ON THE AUDIENCE

Show what they need to understand, not
everything you calculated.

KEEP IT CONSISTENT

Use clear headings, visuals, definitions, and
concise explanations.

EXAMPLE / APPLICATION

Possible outputs: report • dashboard •
presentation • research poster • notebook
• written summary

ANALYST / PROJECT TIP

The final format may change, but the logic of
the project should remain clear.

16 DOCUMENTATION

Can another person understand and reproduce the work?

16

KEY IDEA

Can another person understand and reproduce the work?

ASK → INVESTIGATE

RECORD THE SOURCE

Dataset/source, access date, version, and relevant context.

RECORD THE PROCESS

Cleaning, formulas, queries, transformations, assumptions, decisions.

RECORD LIMITATIONS

Missing information, scope, assumptions, uncertainty, and constraints.

EXAMPLE / APPLICATION

Keep a simple project log: Source → Changes → Methods → Assumptions → Results → Limitations.

ANALYST / PROJECT TIP

Good documentation makes your work easier to review, explain, reproduce, and improve.

17 THE SIMPLE FORMULA

A compact framework students can remember and apply.



THE GOLDEN RULE

Every analysis, visual, finding, and conclusion should connect back to the original project problem.

Start with a question. Use evidence. Explain what it means. Finish with a clear, defensible conclusion.